

MetroPulse — Executive Memo

Urban Mobility Intelligence | NYC | April 1–June 30, 2024

Executive Summary:

MetroPulse looks at **10.59 million NYC Yellow Taxi trips** from **April 1 to June 30, 2024**, along with hourly weather and subway ridership data. The main purpose was to understand how taxi demand changed during the period and identify a few areas that could be tested through small operational pilots.

Key Findings:-

- **Demand:** May had the highest number of trips at 3.66M, compared with 3.46M in April and 3.48M in June.
- **Peak period:** The 5 PM–8 PM period accounted for 2.71M trips (25.62%).
- **Geographic concentration:** Higher trip demand and revenue were mainly seen around Midtown, Upper East Side, and the airport zones.
- **Weather:** Rainy hours averaged approximately **5,524 trips/hour**, compared with **4,725 trips/hour** during no-rain hours. This shows an association, but it does not prove that rain caused the increase.
- **Data quality:** The final fact table matched the cleaned taxi dataset, and all 10 automated quality tests passed.
- **Fare stability:** The average charged amount in April and May differed by approximately \$0.90, with a 95% confidence interval of \$0.72–\$1.09.

Recommended 4-Week Pilot:

Initiative 1 — Peak-Hour Capacity Planning

Use the 5 PM–8 PM demand pattern to test whether more proactive capacity planning can help during the busiest hours.

- Baseline: 2.71M peak-period trips
- Scenario range: a 2%–5% planning scenario corresponds to approximately 54K–136K trips across the observed peak-period volume.
- Track: hourly demand, completed trips, median trip duration, and average charged amount.
- Risk: demand patterns could be different outside the study period.

Initiative 2 — Weather-Responsive Capacity Planning

Test whether using weather-related demand patterns can help with short-term capacity planning.

- Observed rain-hour demand: **5,524 trips/hour**
- Observed no-rain demand: **4,725 trips/hour**
- Scenario range: a 2%–5% planning scenario corresponds to approximately 38K–95K trips across the observed rain-hour volume.
- Track: hourly demand, trip completion, median duration, and weather conditions.
- Risk: the relationship is observational, so weather should not be treated as the only reason for changes in demand.

Decision Framework:

Run both initiatives for 4 weeks and compare the results with the baseline. Use confidence intervals and sensitivity checks when reviewing the changes. Continue, modify, or stop each initiative based on the operational results rather than assuming that historical correlations prove a cause-and-effect relationship.

Data period: April 1–June 30, 2024

Sources: NYC TLC taxi data, NYC taxi-zone data, Open-Meteo weather data, and MTA subway ridership data.

Dashboard: Deployed MetroPulse application with interactive filters and data-quality monitoring.